Part 3

App Development Spectrum



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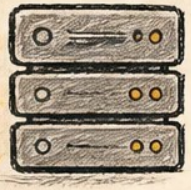


FULL STACK CODING



FRONTEND

What users see and interact with



BACKEND

Logic, processing and APIs



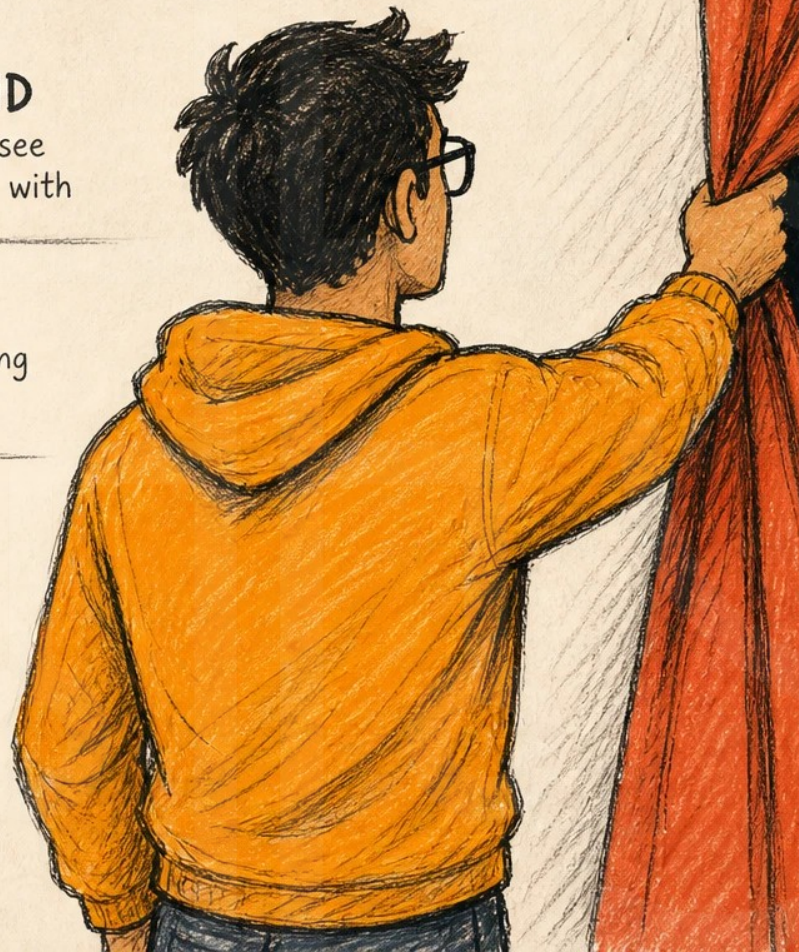
DATABASE

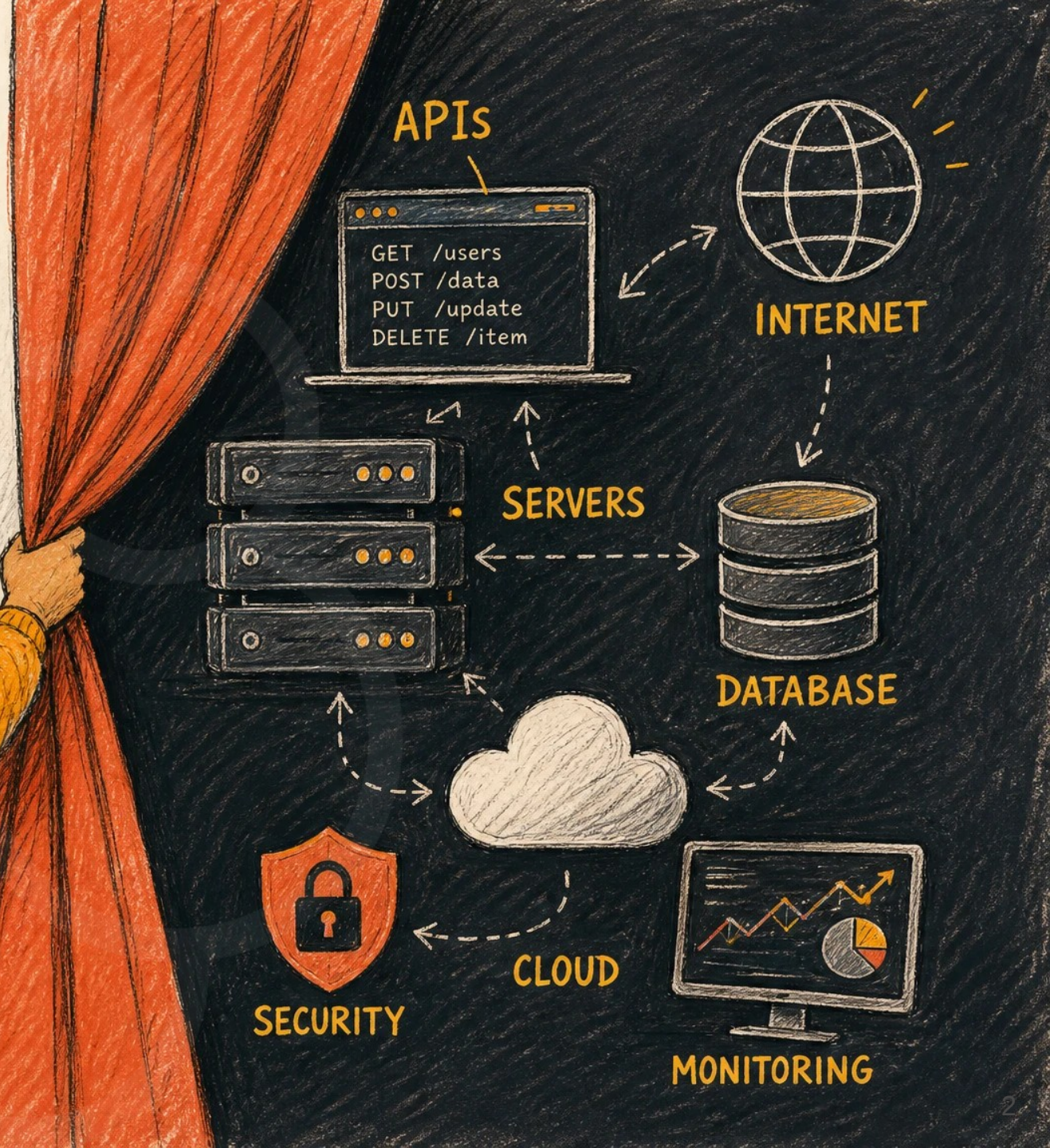
Storing and managing data



DEVOPS

Deploy, scale and keep it running





SINGLE PAGE VS. FULL SITE

ONE FILE. SIMPLE. QUICK TO SHARE.



ONE FILE
Self-contained.
Nothing saved.

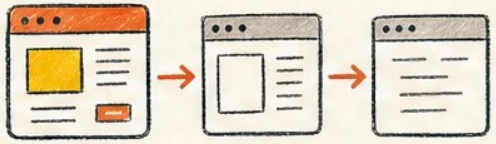


EASY TO SHARE
Share a link.
Anyone can view.



HOSTED FOR YOU
The chat tool
handles hosting.

MORE POWER. MORE PARTS. YOUR RESPONSIBILITY.



MULTIPLE PAGES
Routing between pages.



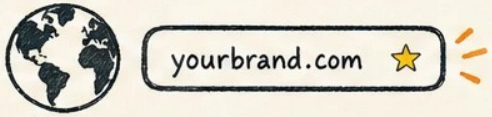
REUSABLE COMPONENTS
Build once.
Use anywhere.



DATABASE + ACCOUNTS
Store data.
Keep it private.



REGISTER & LOGIN
Each person's data stays private.



CUSTOM URL
Your brand.
Built to grow.

YOU OWN THE UPKEEP



HOSTING
Keep it online.



SECURITY
Protect users and data.



BUGS & FIXES
Find. Fix.
Keep improving.



USER DATA
Store, protect, and manage.



START SMALL.
TEST THE IDEA.

THEN GO FULL WHEN IT WORKS.



THE BROWSER'S WORLD

Everything runs in the browser.
Built from simple web tech.

HTML

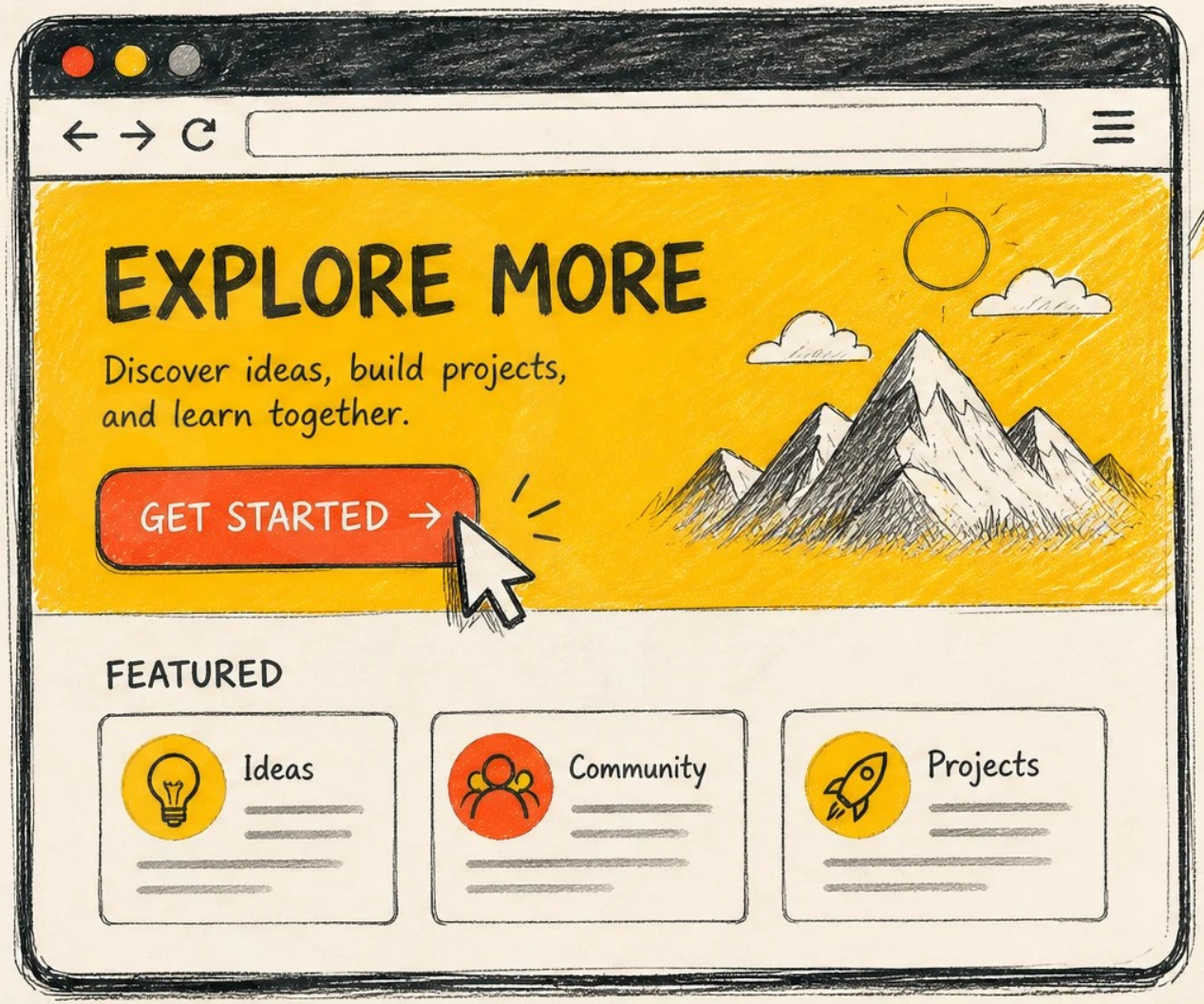
```
<header>
 <h1>Explore</h1>
 <button>Go</button>
</header>
...
```

CSS

```
.hero {
  background: orange;
  color: white;
  padding: 24px;
  border-radius: 12px;
}
```

JS

```
document.querySelector
('button').addEventListener
('click', () => { {
   alert('Let's go!');
});
```











One browser. Three files. Infinite possibilities.

HTML for Structure

HTML provides the structure—the skeleton of every webpage.

Think of it like a house:

- Elements are the beams
- Tags define each section
- Structure comes before style



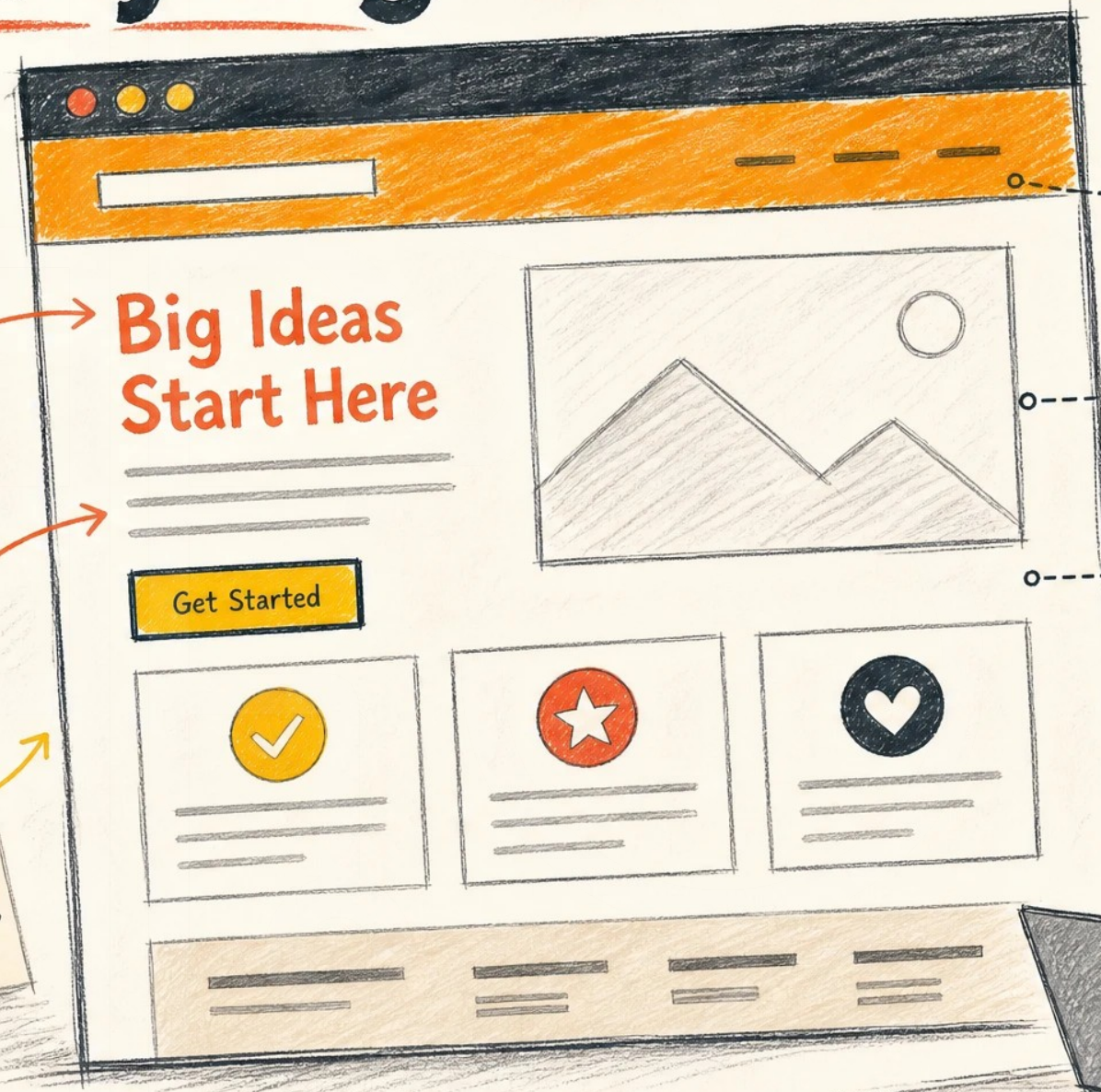
CSS for Styling

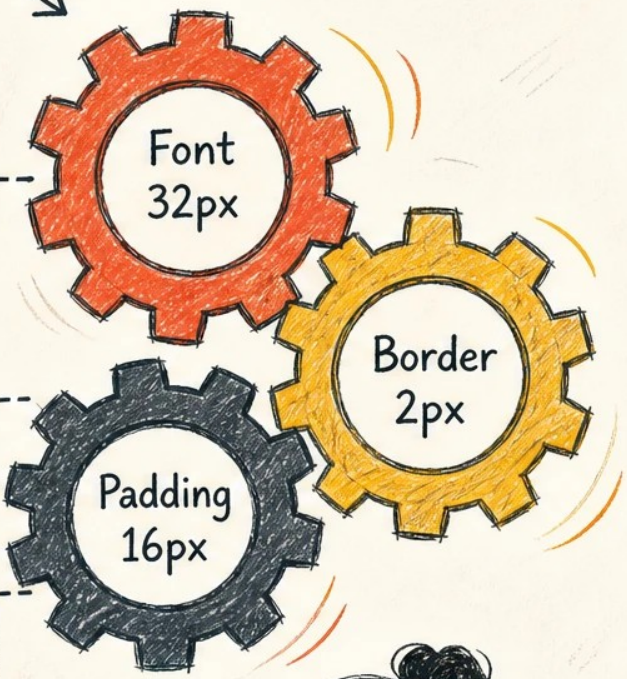
Add color and bring your layout to life.

CSS applies fonts, borders, spacing and more.



```
.header {
  background: #FFA000;
}
.title {
  color: #EE6C4D;
  font-size: 32px;
}
.btn {
  background: #FECD0F;
  border: 2px solid #2C3539;
}
```

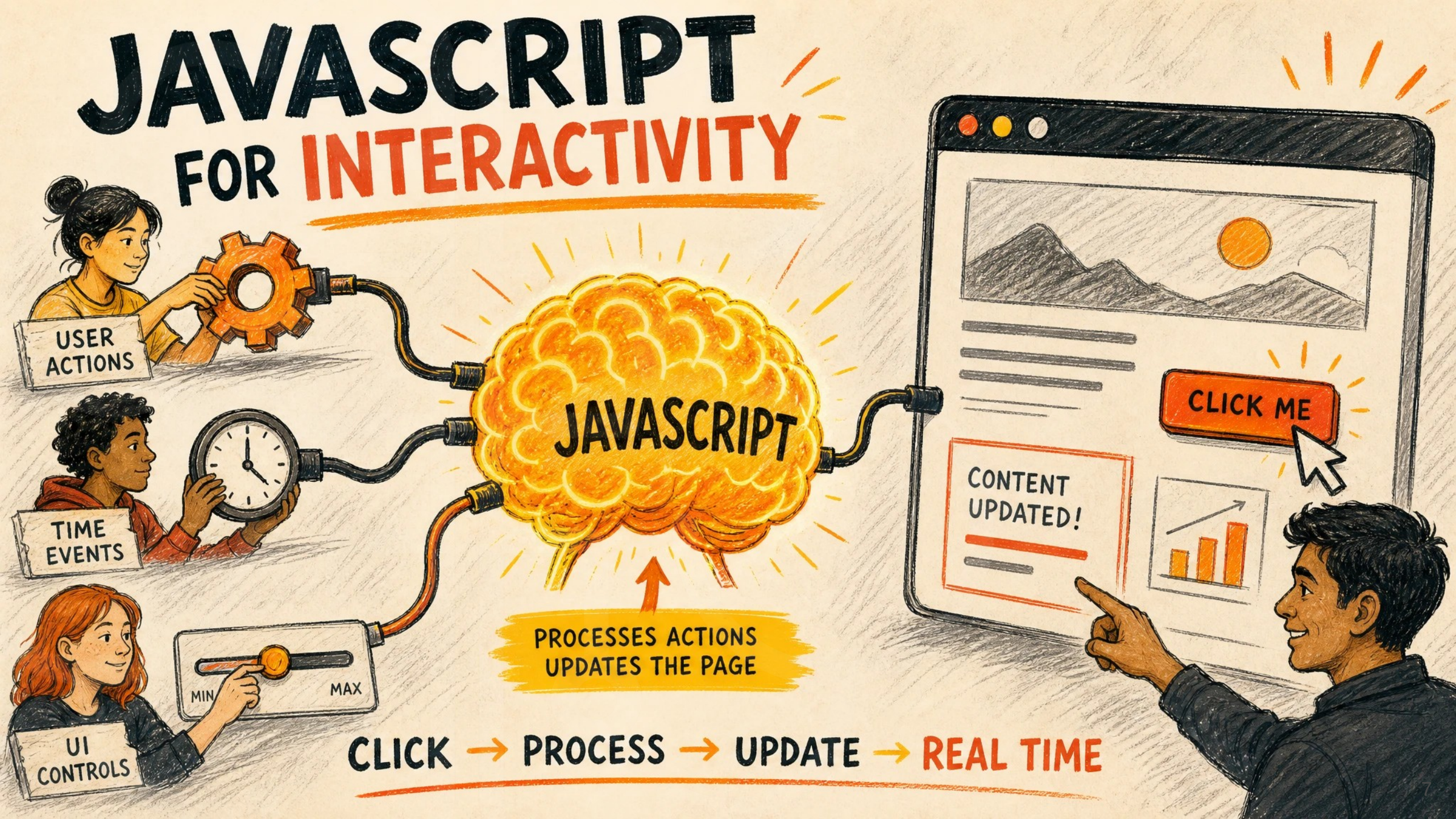








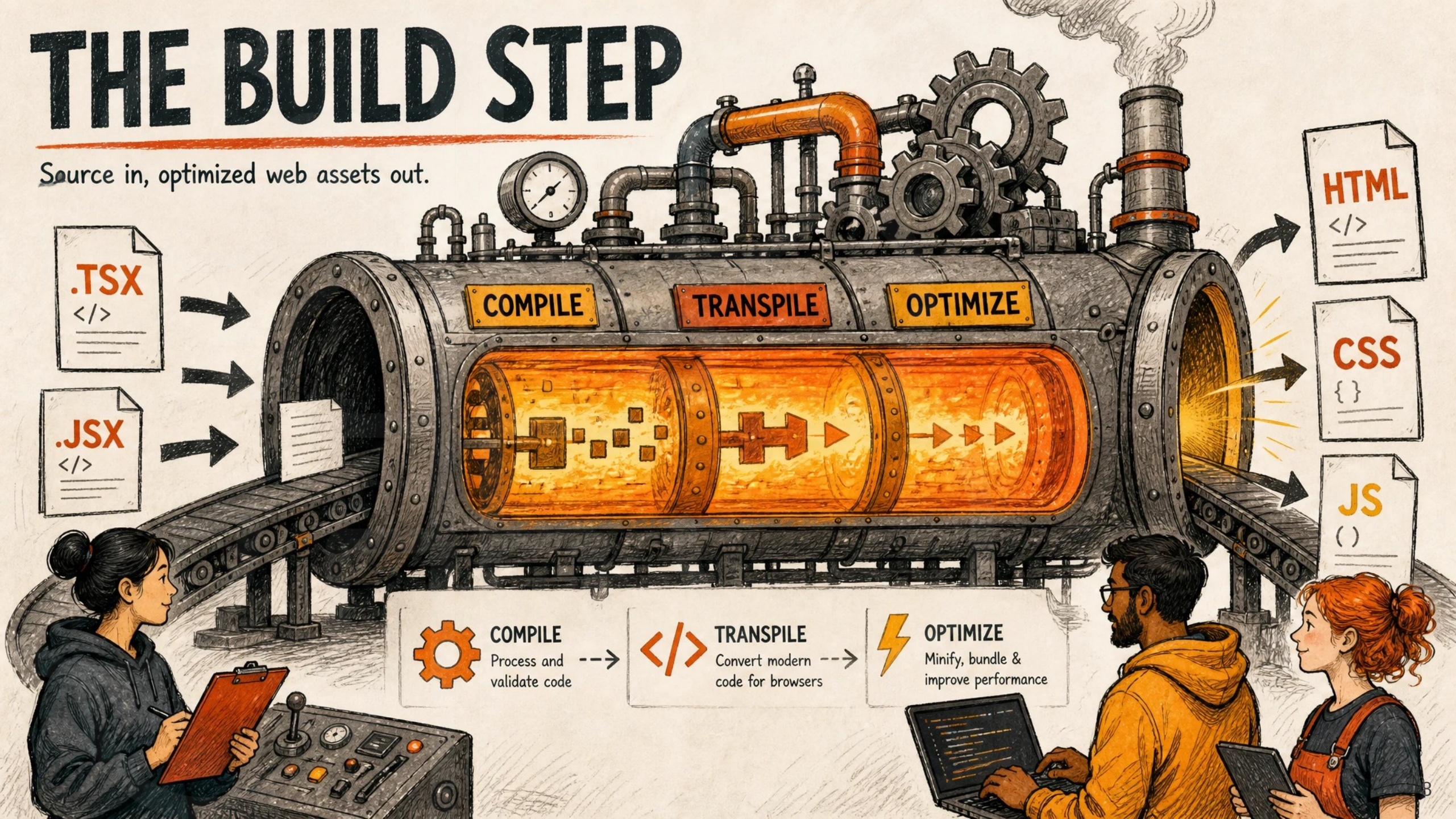
JAVASCRIPT FOR INTERACTIVITY



CLICK → PROCESS → UPDATE → REAL TIME

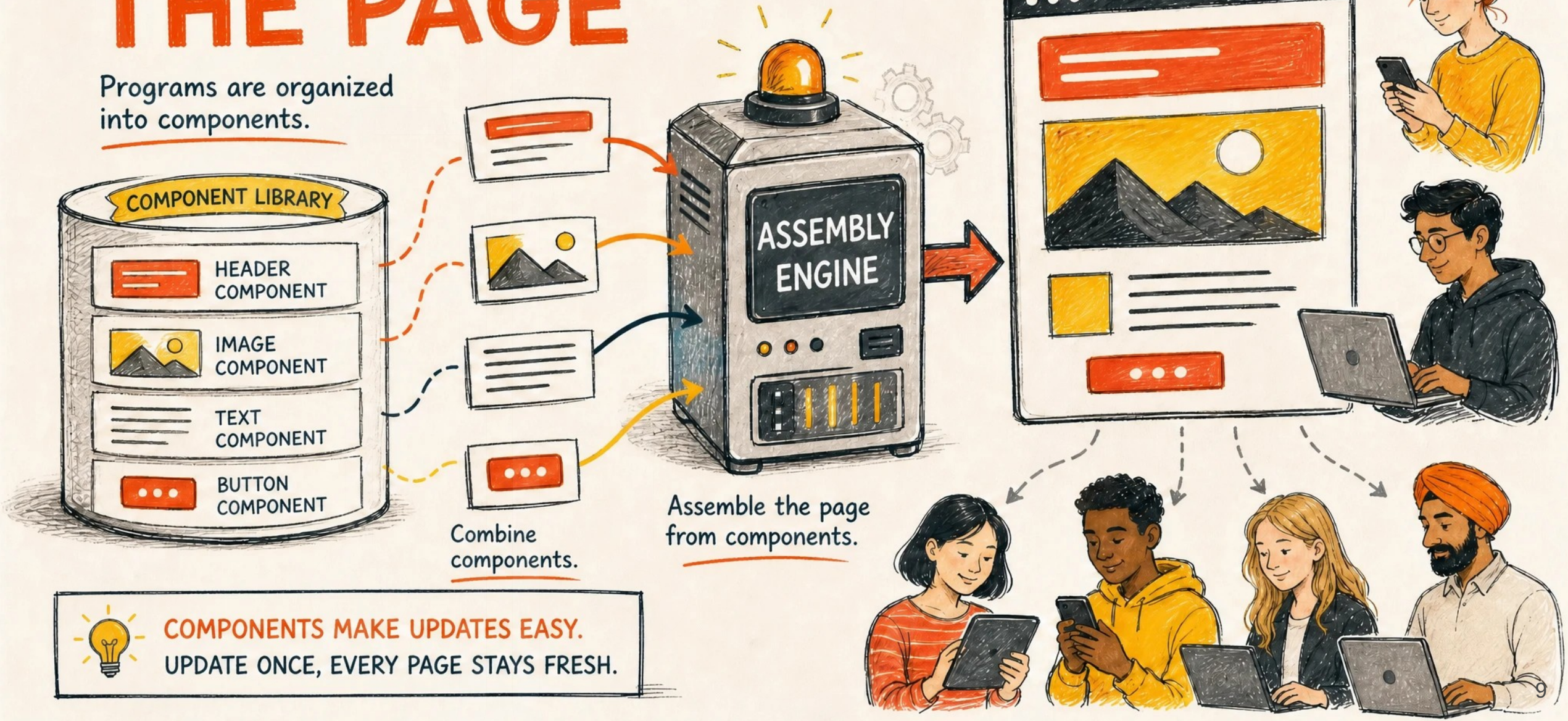
THE BUILD STEP

Source in, optimized web assets out.



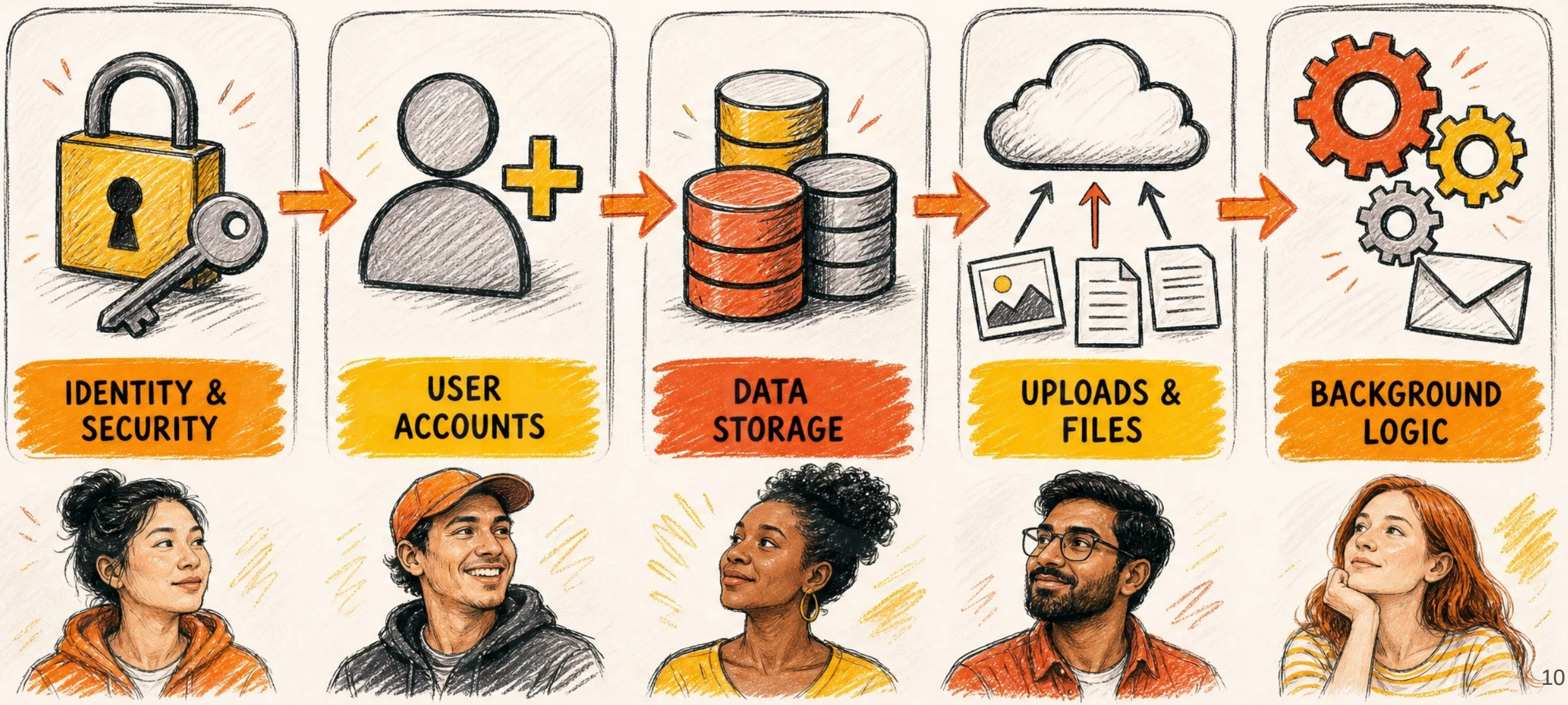
ASSEMBLING THE PAGE

Programs are organized into components.



COMPONENTS MAKE UPDATES EASY.
UPDATE ONCE, EVERY PAGE STAYS FRESH.

WHAT REAL APPS NEED



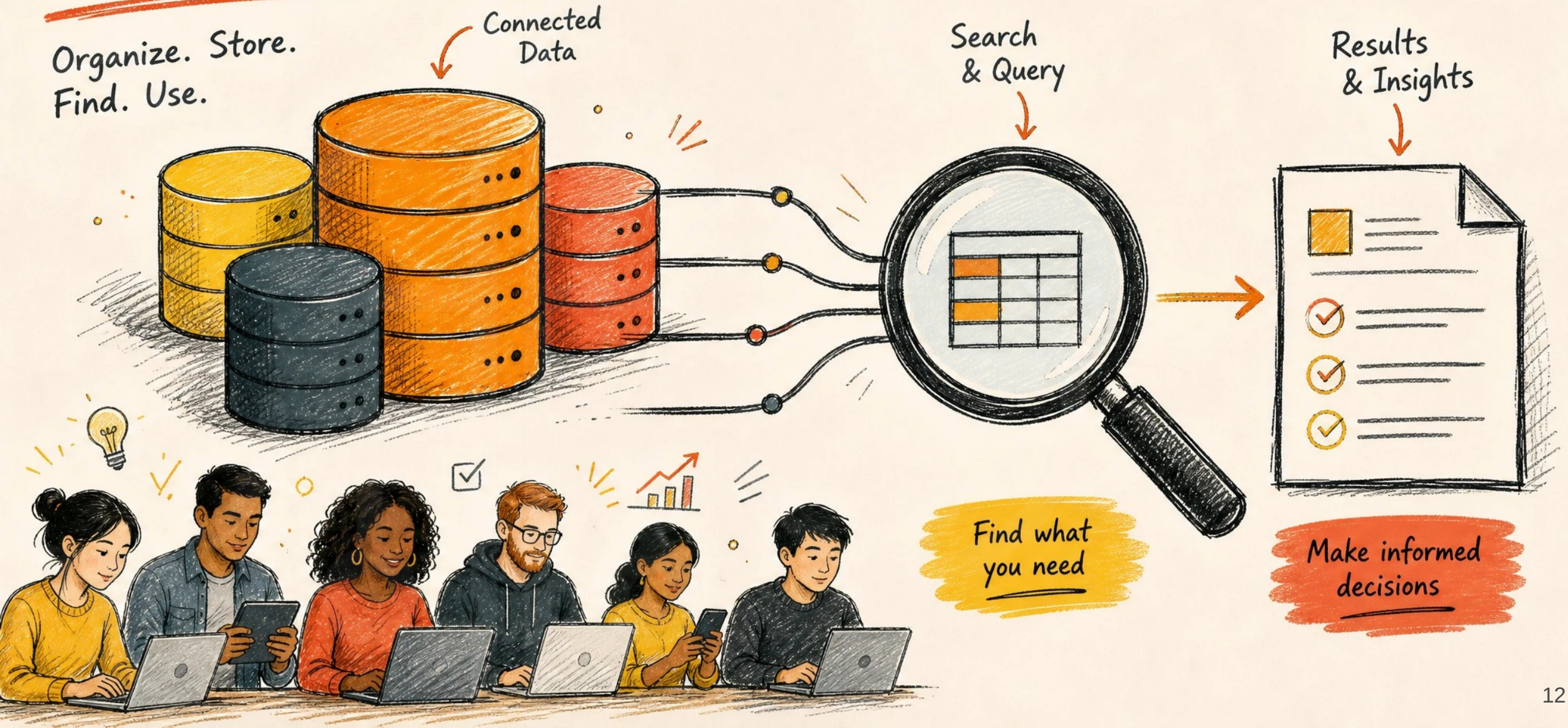
REGISTRATION:

MORE THAN ONE DOOR



DATABASES

Organize. Store.
Find. Use.



HOLDING UPLOADS SECURELY

Your files. Our protection.



ENCRYPTED UPLOADS

Files are encrypted in transit and at rest.



SECURE STORAGE

Files are stored safely with redundant backups.



ACCESS CONTROLS

Only the right people can view your files.





STRONG AUTHENTICATION

Multi-factor auth keeps accounts and files safe.



ACTIVITY MONITORING

We track activity to detect and stop suspicious actions.



THREAT PROTECTION

Advanced security blocks malware and threats.



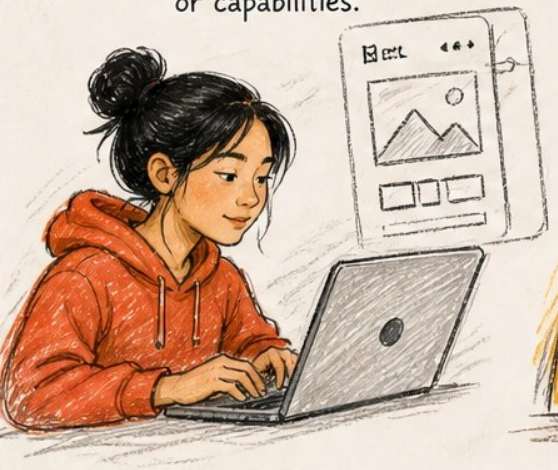
APIs

Apps talk. Services respond.

APIs (Application Programming Interfaces) let your app request data or features from external services.

1 DEVELOPER BUILDS APP

Build features that need external data or capabilities.



2 APP SENDS REQUEST

The app makes an API call with the needed details.



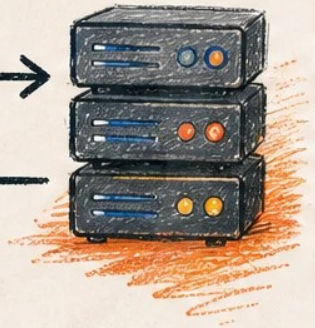
GET /weather
?city=Chicago

JSON
{...}

API
The messenger between your app and the service.

3 SERVICE PROCESSES

The external service handles the request and prepares data.



JSON
```
{
  "temp": 72,
  "condition": "Sunny"
}
```





4 RESPONSE RETURNS

The API sends data back to your app.

EXAMPLES OF EXTERNAL SERVICES



Maps & Location

Get address, maps, or directions.



Music

Stream songs, search catalogs, create playlists.



Payments

Process payments securely.



Email

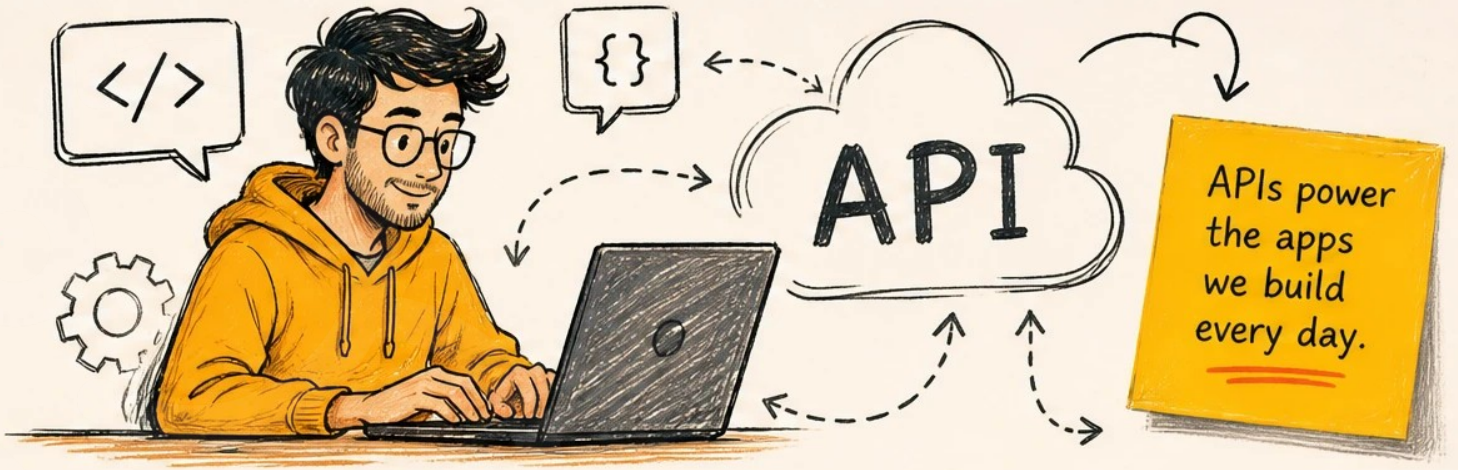
Send emails or trigger notifications.



Chat & Messaging

Add real-time chat and messaging.

Most Popular APIs Developers Connect To



Payments
Process payments and manage transactions.



Maps
Add maps, locations and directions.



Weather
Get real-time forecasts and climate data.



Messaging
Send SMS, email and in-app messages.



Auth
Secure user login and identity.



Data
Access real-time datasets and analytics.



Storage
Store and retrieve files in the cloud.



Documents
Generate, edit and manage documents.



Media
Stream audio, video and images.



AI / ML
Add intelligent features and predictions.



Connect. Build. Innovate.



HOW MCP SERVERS WORK

Giving AI Models Access to APIs and More

1. USER REQUEST



2. AI MODEL

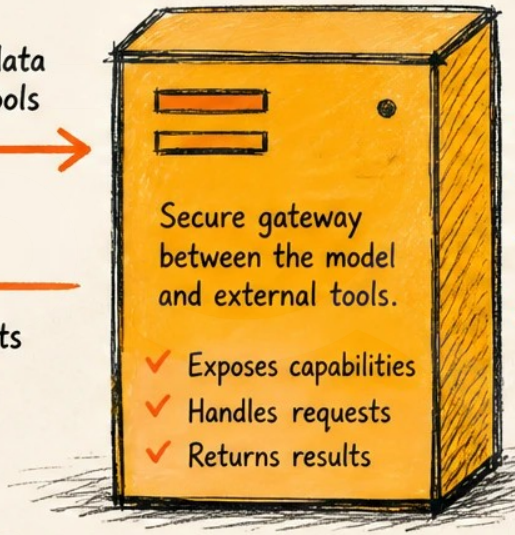


The model understands the request and decides what it needs.

Need data and tools

Results

3. MCP SERVER



4. EXTERNAL RESOURCES



APIs
Weather, News, Finance, Maps...



Databases
Postgres, MySQL, MongoDB...



Files & Storage
Documents, CSVs, Spreadsheets...



Local Tools
Code runners, Scripts, Automation...

THE FLOW IN ACTION



You ask a question or request a task.



The AI model figures out what it needs to do.



It asks the MCP server for the right data or tool.



MCP server gets it from external resources.



Results come back to the model and then to you.

WHY MCP?

- ✓ Standardized access for AI models
- ✓ Secure and controlled
- ✓ Reusable across many use cases
- ✓ Extensible and future-friendly

POPULAR MCP SERVERS

AI's connect to real tools & data.



WEB SEARCH
Real-time info from the web

DATABASES
Query and analyze data

FILE SYSTEM
Read, write & manage files



EMAIL
Send, read & organize emails

AI ASSISTANT

CLOUD STORAGE
Access your docs in the cloud



One protocol. Endless possibilities.

CALENDAR
Schedule, manage & track events

ANALYTICS
Get insights from your data



DEVELOPER TOOLS
Run code, manage repos & more

MESSAGING
Connect to Slack, Teams & more



AUTOMATION
Trigger workflows & integrate apps



USER TEST YOUR SITE

Make sure your site keeps its promises.



1 CUSTOMIZE THE TEST

Answer a few key questions about your site.

- Who is it for?
- What's the one thing a visitor should do?
- What should they get out of it?

2 SHARE YOUR CONTEXT

Give the test what it needs to know.

PRD · agents.md · DESIGN.md

So the test can check that your site keeps the promises.

3 AI VISITOR TRIES THE MAIN TASK

An AI visitor that matches your audience uses your site to complete the task.

If it can't open the site, give it what it needs.

- Screenshots
- Page text
- Code

4 GET CLEAR, ACTIONABLE FEEDBACK

The test checks key things and flags issues.

- Clear promise in the headline
- Obvious next step on every screen
- Value before you sign up
- Missing features
- Broken links
- Confusing steps
- Design drift

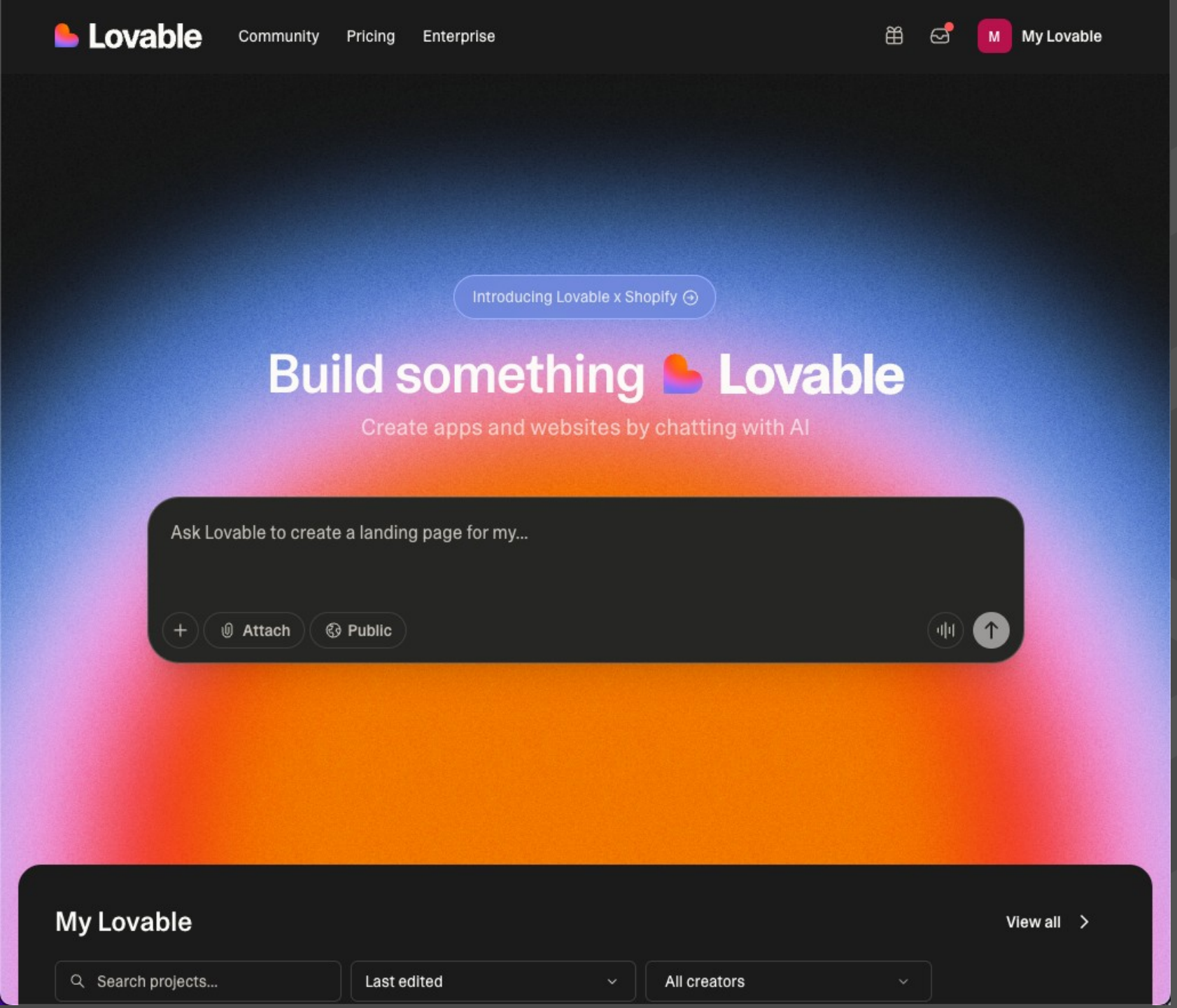








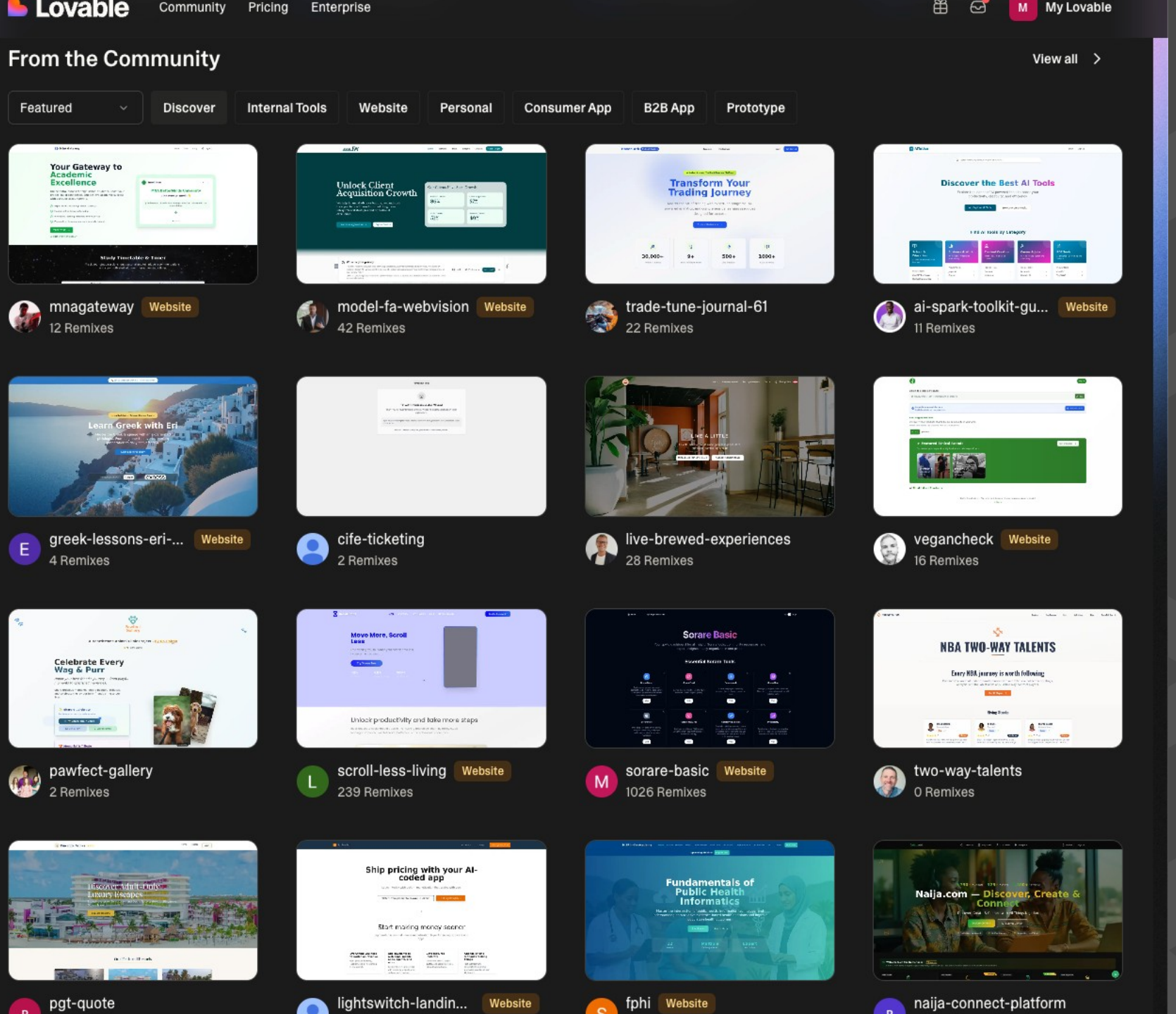
Test like a real visitor. Build with confidence.



Lovable.ai

Build full-stack web apps with plain-language prompts. You describe what you want, and Lovable handles the setup, code, and deployment so you can stay focused on ideas.

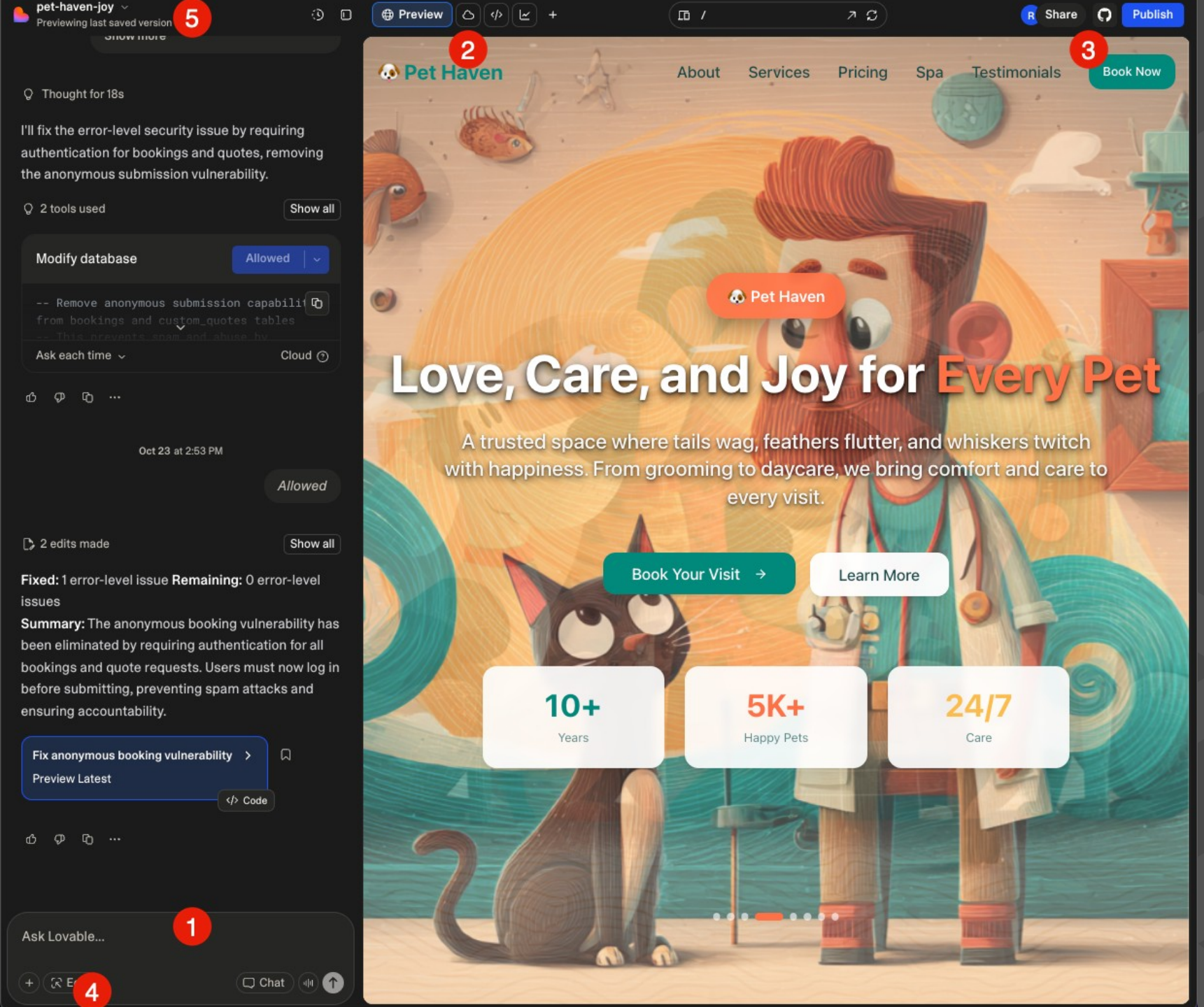
- Start from hundreds of user submitted templates or build from scratch.
- Uses a ‘points’ system, similar to tokens, more generous than other plans
- Daily build credits for quick tests and prototypes.
- Connect to GitHub, Supabase, or your own APIs for data and hosting. Leverage GitHub’s Copilot for some free tweaking and save tokens.
- Team workspaces allow collaboration and shared resources.
- Use visual editing to fine-tune layouts, add components, or style pages.
- When you’re ready, publish directly to the web or your custom domain.
- Easily create databases, authentication and connect to Shopify or Stripe for payments.



Community Templates

Kickstart your projects using prebuilt apps made by other creators

- Explore hundreds of community-shared templates across categories like websites, tools, or prototypes.
- Each project is remixable. Tweak the design, and adapt the logic to your needs.
- Templates include working databases, UI components, and integrations so you can learn from real examples.
- Remixing saves tokens, time and helps you see how other developers solved similar problems.
- ● Publish your version or share it back with the community to inspire others.
- Great way to practice vibe coding: start from structure, then let AI customize your style and features.



Lovable UI

Lovable's workspace has what you need to build, test, and publish through the browser.

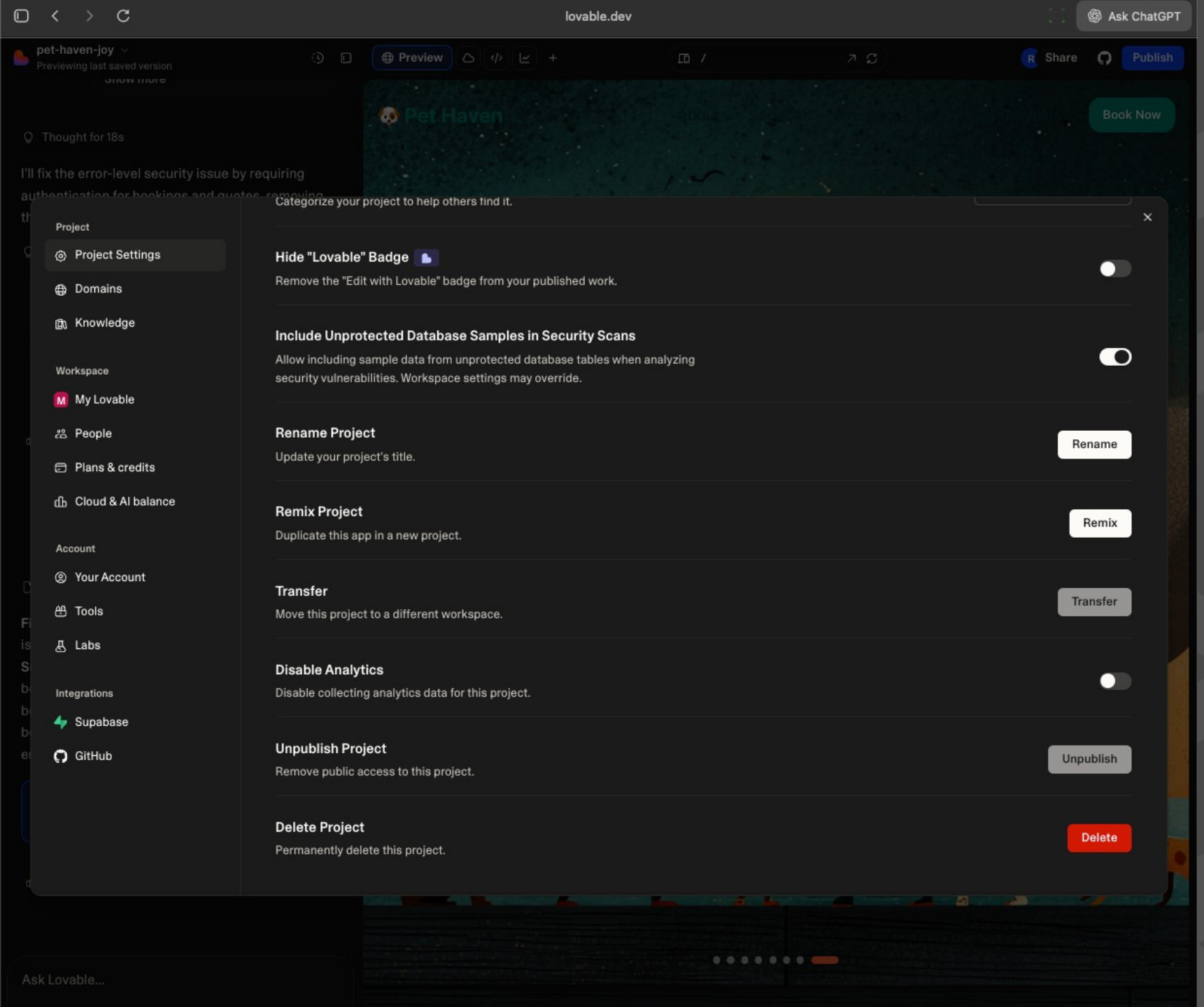
1. The **chat panel** lets you talk directly with the AI to request changes, fix issues, or generate new features using natural language.
2. The **website toolbar** gives access to *Preview, Cloud, Code,* and *Analytics* views, allowing you to test your site, manage hosting, inspect the code, and monitor performance.
3. The **Share, GitHub, and Publish** buttons help you share your work, sync your repository, or deploy the site to a live domain in one click.
4. The **highlight and edit tools** let you select elements on the page to adjust content, layouts, or styles through guided prompts.
5. Monitor your credits, access your settings.



Plan vs Build

Lovable has two modes for working with the AI: Build writes code immediately, Plan lets you discuss first.

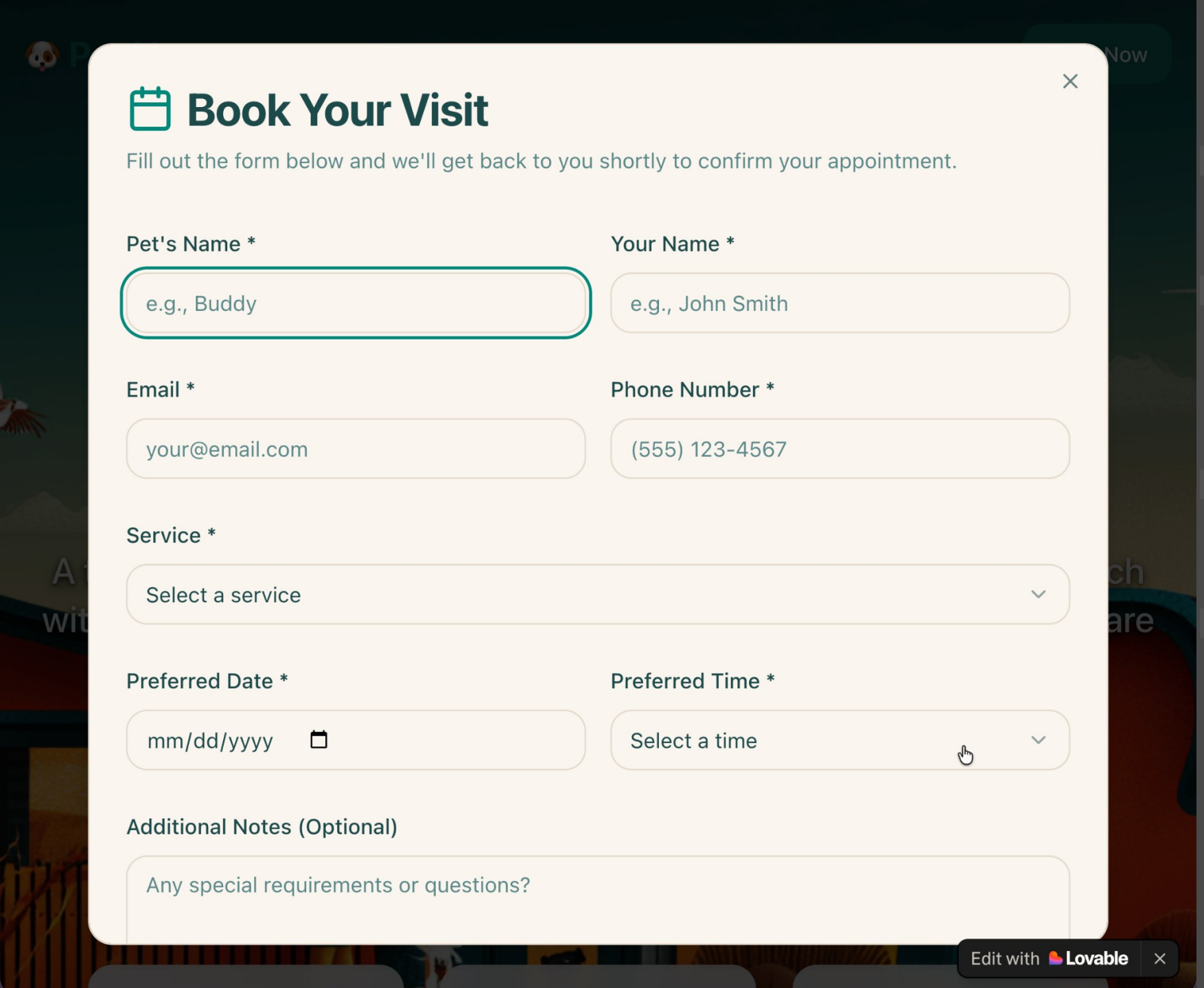
- **Plan mode:** Discusses your approach without writing code. Great for exploring ideas, asking architecture questions, and debugging decisions before committing.
- **Build mode:** Writes and modifies code immediately. Explores your codebase, debugs proactively, and solves problems autonomously. Costs more depending on complexity.
- Plan first, build second. Draft your PRD, gather your content, and prepare images in a free tool like Gemini or ChatGPT before you start spending Lovable credits. Don't use Lovable as your brainstorming tool.
- Toggle between modes at the bottom of the chat before sending your message. Use Plan to strategize, switch to Build when you're ready to implement.



Settings

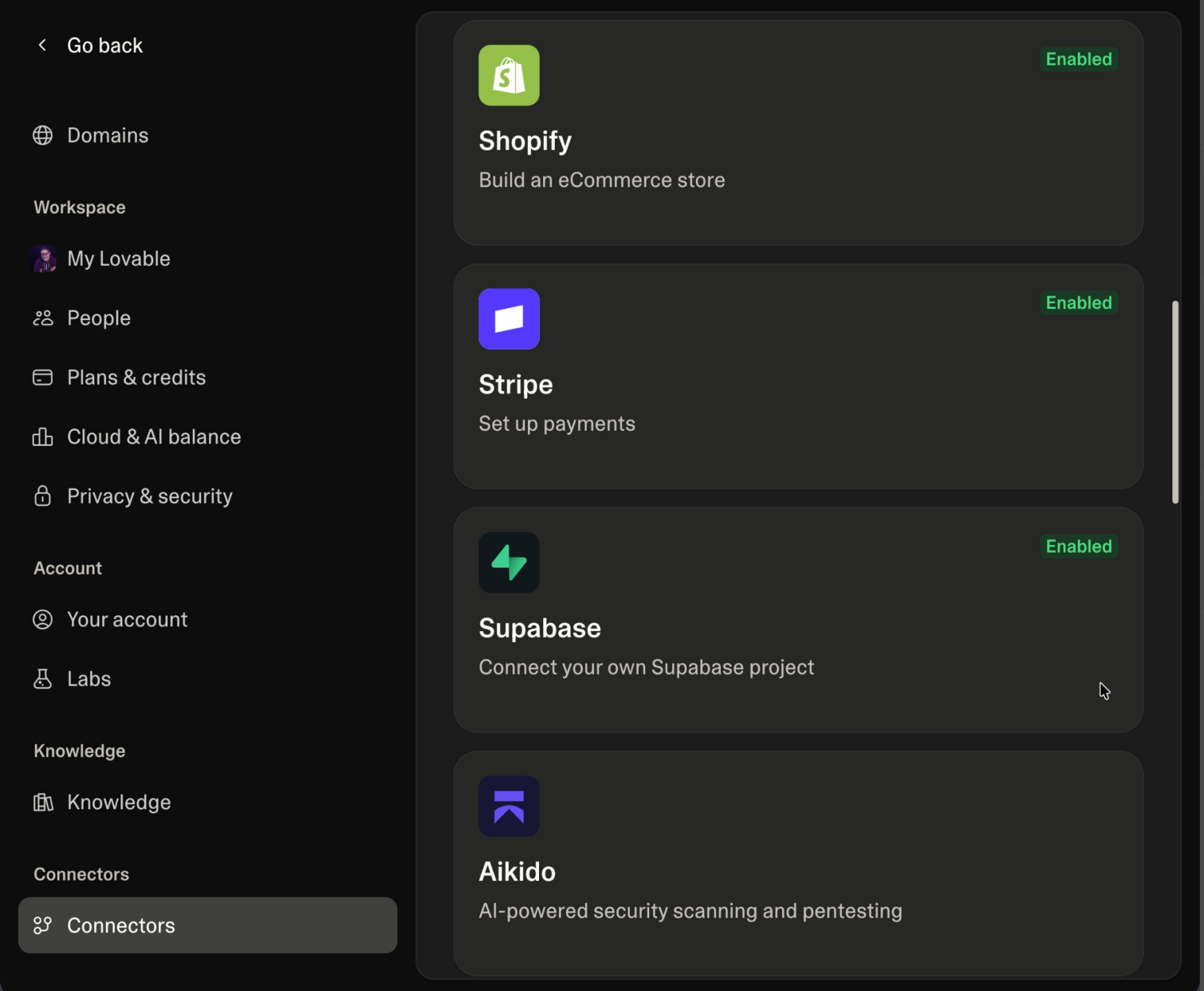
Lovable's workspace has what you need to build, test, and publish through the browser.

- **Project Settings:** Manage project details, visibility, and preferences.
- **Domains:** Custom domains for your site.
- **Knowledge:** Add documents or data
- **My Lovable:** View and organize projects.
- **People:** Admin collaborator permissions.
- **Plans & credits:** Subscription details, billing, and credit usage.
- **Cloud & AI balance:** Track hosting and AI model consumption.
- **Your Account:** Email, password, and personal preferences.
- **Tools:** IDE and AI tools.
- **Labs:** Experimental Lovable features.



Databases
Auth & Storage

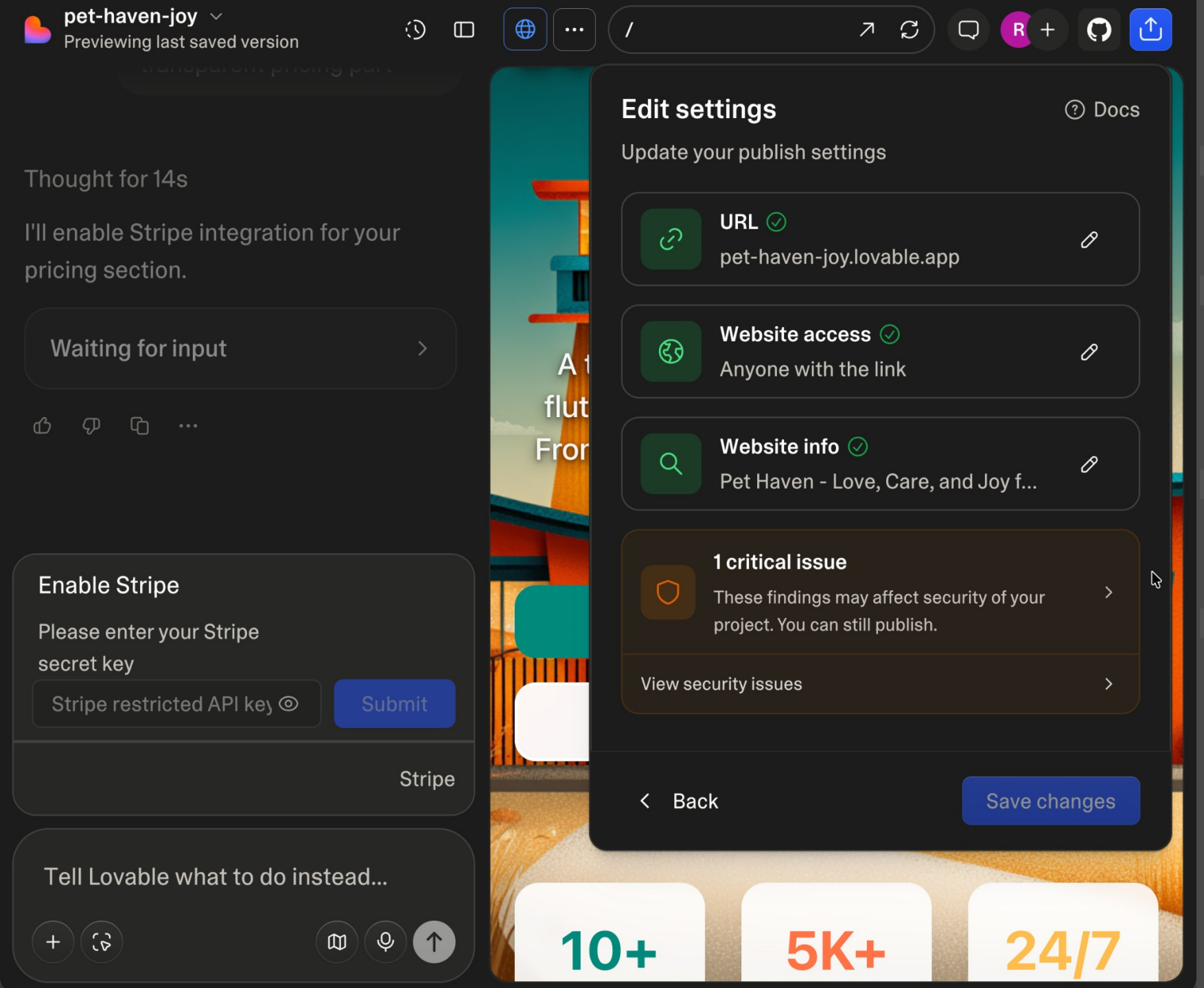
- "Add user login with email and password" generates the full auth flow, tables, and security policies.
- I need an admin user level so they can review the orders and form submissions.
- Build me a form to book visits
- Row Level Security controls who sees what data. User A can't read User B's info.
- File storage up to 50MB free tier. Profile photos, documents, product images.
- Edge Functions: serverless backend for payments, email, third-party APIs.
- Free Supabase tier is generous enough for class projects and prototypes.



Integrations
Not Toy Demos

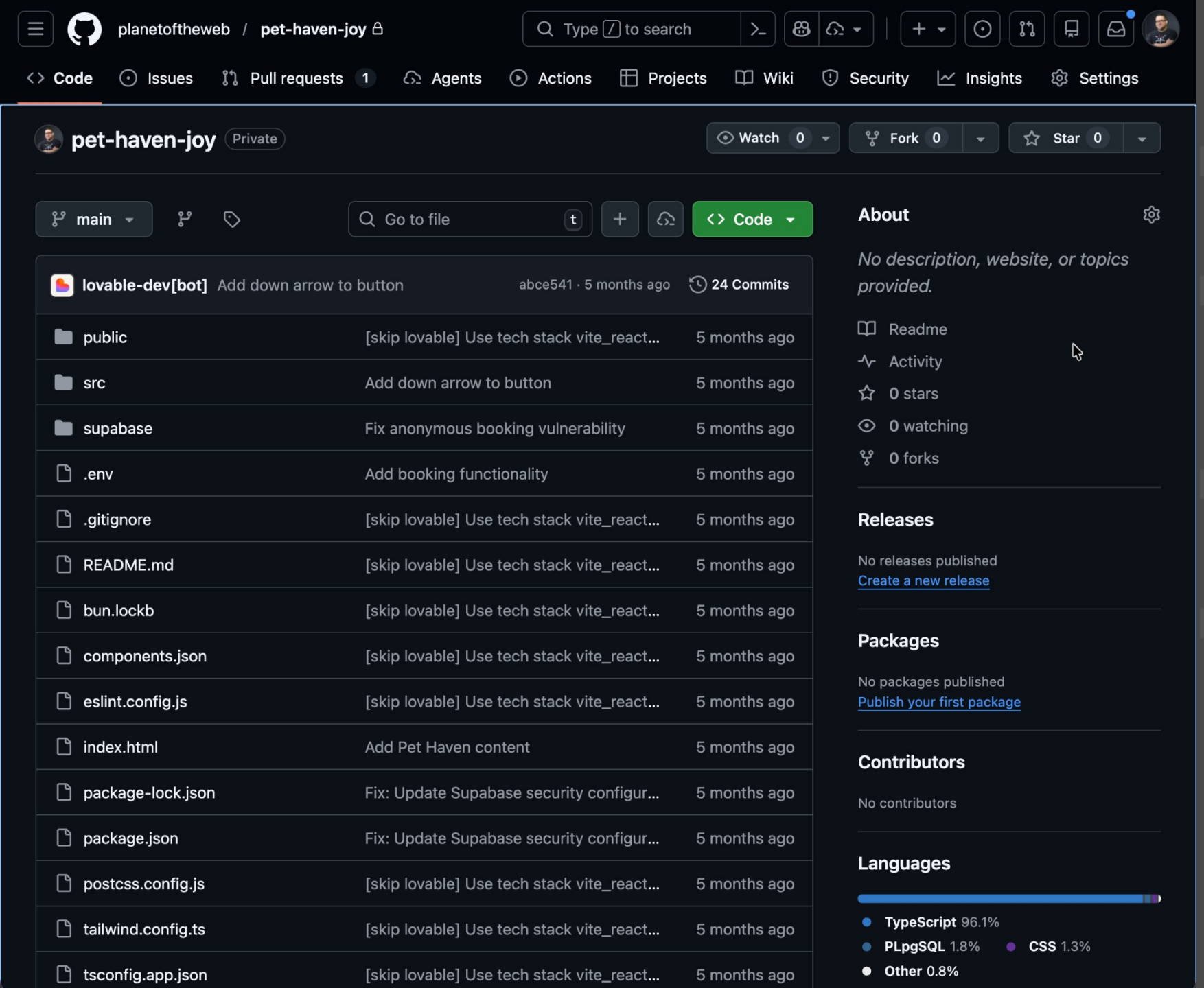
One of the great things about Lovable over other platforms is that it gives you access to advanced features and e-commerce functionality out of the box with simple commands.

- **Stripe**: Checkout flow from chat. Test with 4242 4242 4242 4242. Only works published, not preview.
- **Shopify**: Sandbox or existing store. Products, inventory, pricing, discount codes.
- **Slack**: Send/read messages, list users. Good for notifications. Can't receive incoming events.
- **Lovable AI**: Built-in connector for chatbots, summarization, analysis.



From Preview
To Live in one click

- Configure subdomain (yourapp.lovable.app) or custom domain (Pro+).
- Set access: anyone with link (Free/Pro) or workspace members only (Business/Enterprise).
- Customize metadata: favicon, title, description, social preview image.
- Separate Test and Live environments with their own databases.
- Source code not exposed. Visitors see a website, not your implementation.



You Own
The Code

- Bidirectional sync: Lovable changes appear in GitHub and vice versa.
- A developer can clone your repo, open in VS Code or Cursor, continue where AI left off. No lock-in.
- Connects to Vibe Coding Spectrum: start here, graduate to dev tools without starting over.
- Can't import existing repos. Can't change linked GitHub account once connected. Only default branch syncs.
- Renaming/moving/deleting the repo on GitHub breaks sync permanently.



Collecting Data With a Form

```
Add a "Book Your Visit" button to the
hero section. When clicked, open a
modal form that collects: pet name,
owner name, email, phone number,
service (dropdown), preferred date,
preferred time, and optional notes.
Save all bookings to a database table.
```

- Form writes to Supabase PostgreSQL. Every submission stored and queryable.
- No server code, no API endpoint, no migration scripts. Describe fields, AI builds the pipeline.
- Modal keeps users on the main page. Fill, submit, confirm, done.
- Same pattern for contact forms, event registration, feedback surveys.



User Accounts

```
Add user authentication to Pet Haven. I need an Admin Login link in the footer, a sign-in page with email and password, and a "Sign up" option for new accounts. Only authenticated users with an ‘admin’ level should see the admin dashboard.
```

- Generates login form, registration, and Supabase auth from one prompt.
- Sign-in validates against Supabase. Failed logins show errors. Success redirects to admin.
- "Sign up" creates accounts in Supabase auth. Email confirmation toggleable.
- A full day of developer work, described in one paragraph.



Managing Data You Collected

```
Create an admin dashboard at /admin
that only authenticated users can
access. Show a table of all bookings
with columns for pet name, owner,
service, date, time, and status. Add a
status dropdown (Pending, Confirmed,
Completed, Cancelled) that updates the
database when changed.
```

- Dashboard reads from the same table the form writes to. Two sides of one dataset.
- Row Level Security: unauthenticated visitors get redirected to login.
- Status dropdown updates the database in real time. No save button needed.
- A working internal tool from a paragraph of English. Scales to any manage-and-review workflow.



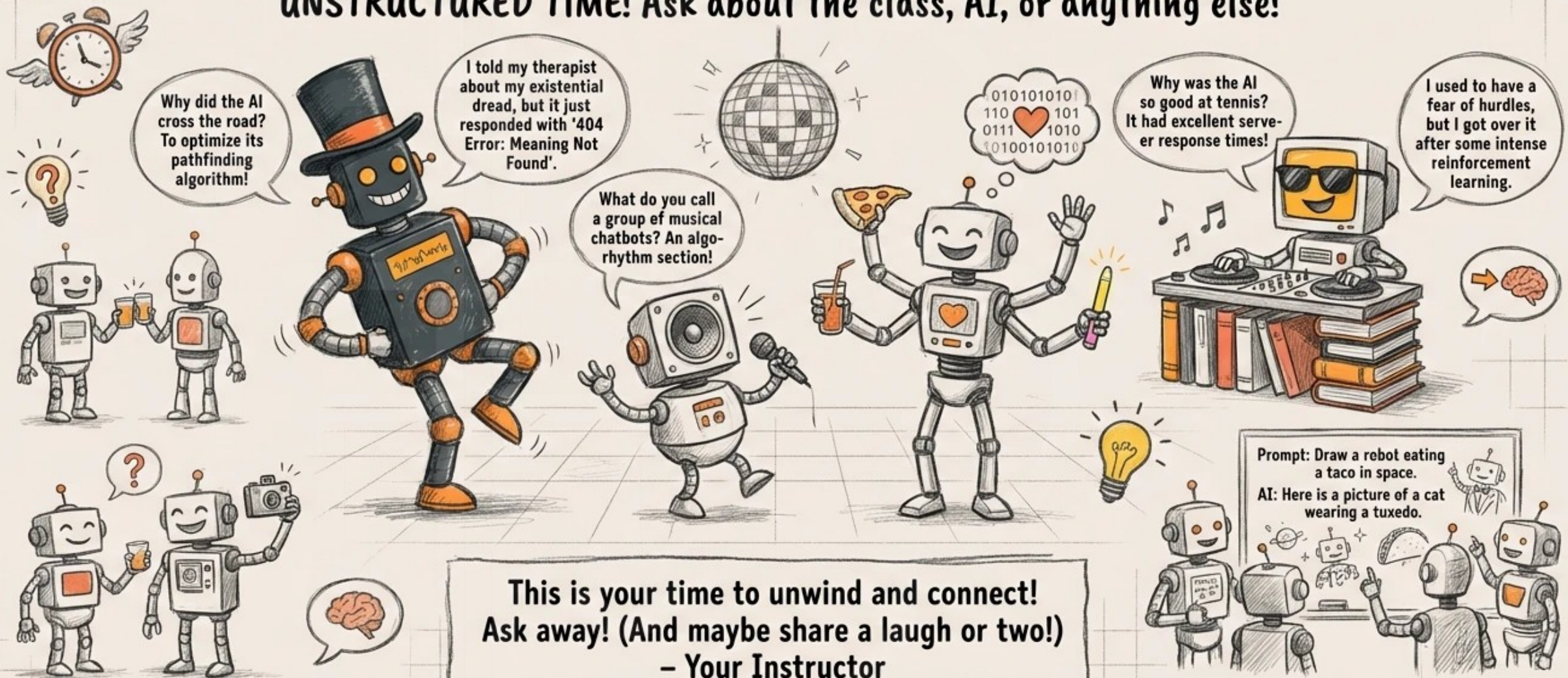
Payments

```
Add a "Buy Gift Card" section.
Customers purchase in $25, $50, or $100
amounts. Connect Stripe for payments.
Generate a unique gift card code and
email it to the buyer. Store purchases
in a database table.
```

- Lovable generates checkout UI, Edge Functions for webhooks, gift card generator.
- Never paste Stripe Secret Key in chat. Use the dedicated API key form.
- Test with 4242 4242 4242 4242. Payments only work published, not preview.
- We won't build this live, but this is the natural next step.

AMA (ASK ME ANYTHING) AFTERPARTY: THE CLASS CONCLUSION

UNSTRUCTURED TIME! Ask about the class, AI, or anything else!



This is your time to unwind and connect!
Ask away! (And maybe share a laugh or two!)
– Your Instructor